Superficial Cancer of the Esophagus and GE Junction

1

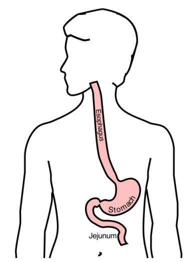
Anatomy

Food moves from the throat

→ esophagus

→ stomach

→ small bowel (jejunum)



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Types of Esophageal Cancer

There are two common types of esophageal cancer

- Adenocarcinoma
- Squamous Cell Carcinoma

In many ways, these to different types of esophageal cancer behave the same.

3

Cancer Staging

Staging refers to the tests to determine

- How large is the tumor?
- Has there been spread to lymph nodes?
- Has it spread to other parts of the body?

Treatment options depend upon the cancer stage

4

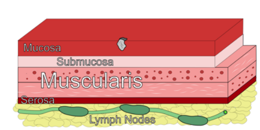
Cancer Staging

- **T** = Tumor - Depth of growth into the wall
- **N** = Nodes - Spread to the lymph nodes
- **M** = Metastasis - Spread to liver, lungs, or bone

5

Early Stage Cancers

Cancers start on the very inside layer called the mucosa



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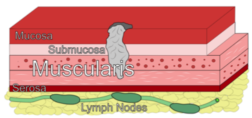
Locally-advanced Cancers

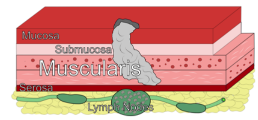
Over time, cancers can grow into the muscular wall



7

Lymph Nodes

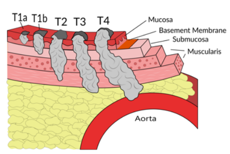
In some cases, cancer cells can break off from the main tumor and spread to lymph nodes



8

T Stage

Cancers are categorized based upon the thickness of the tumor, known as the T stage

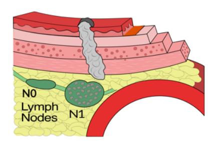


9

N Stage

Cancers are categorized by whether there is spread to the nodes.

- **N0** cancers have not spread to the nodes
- **N1** cancers have spread to the nodes.

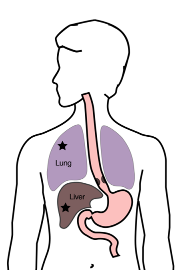


10

M Stage

Some cancers spread to other parts of the body

- **M0** cancers have not spread
- **M1** cancers have spread to lungs, liver, bone, or within the abdomen



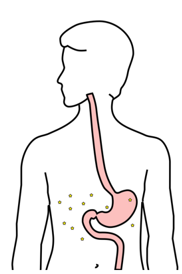
11

Carcinomatosis (M1)

Spread within the abdomen is **carcinomatosis**

Tumors can be very small (grain of rice)

Fluid can also build up: **ascites**



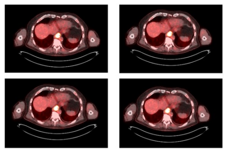
12

PET scan

Similar to CT scan

Tracer shows 'hot spots'

- Cancer
- Inflammation or infection
- Normal organs (heart, kidneys)



13

PET scan Preparation

- Nothing to eat or drink for 4 hours prior to exam
- High-protein low-carbohydrate diet the day prior
- No strenous activity day prior to study

14

PET scan Prepration - Diabetic Patients

PET scan uses a modified sugar as a tracer ⇒ you will need to be careful about blood sugar and insulin before the study

- No insulin within 4 hours of the exam
- OK to take diabetes pills the day of the exam
- PET scans are generally best in the morning
- Plan to eat a meal after the PET scan, with half the usual morning dose of insulin

15

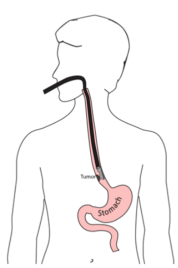
PET scan Prepration - Diabetic Patients

- If PET is scheduled after 10AM:
  - low-carbohydrate breakfast at least 4 hours before
  - half the usual regular (short-acting) insulin
- No long-acting insulin after midnight.
- Insulin pump: turn it off 4 hours before the study (or turn it down to basal/night-time rate)

16

Endoscopic Ultrasound

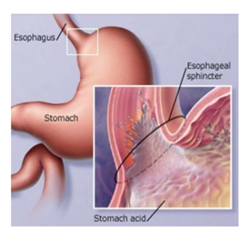
- Similar to upper endoscopy (EGD)
- Ultrasound in scope
- Evaluates T stage



17

Gastroesophageal Reflux

A one-way valve normally keeps acid within the stomach and prevents it from entering the esophagus

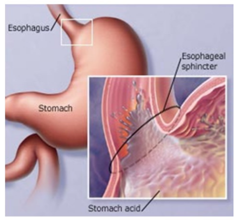


18

Gastroesophageal Reflux

A one-way valve normally keeps acid within the stomach

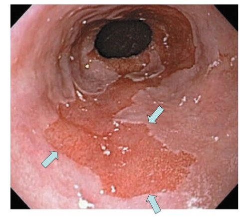
If the valve does not work properly, acid enters the esophagus and cause heartburn and damage to the lining of the esophagus.



19

Barrett's Esophagus

Over time, the lining of the esophagus undergoes change in response to the acid.



20

Dysplasia

Over a period of years, pre-cancerous changes can develop within Barrett's esopahgus.

These changes can be seen by the pathologist from biopsies taken from the esophagus

Over time, low-grade dysplasia can progress to high-grade dysplasia

21

Dysplasia → Cancer

Low grade dysplasia: Risk of cancer 0.5% per year

High-grade dysplasia: Risk of cancer 5% per year

⇒ Surveillance with upper endoscopy is critical

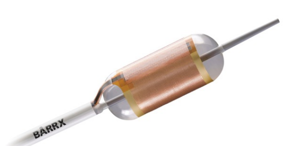
22

Radiofrequency ablation for Dysplasia

Dysplasia can be treated with destroying the mucosa, the inner layer of esophgus

Ablation of the mucosa with microwave energy

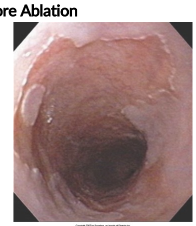
Circular balloon with an antenna used to ablate the mucosa



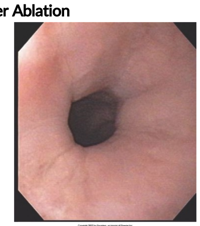
23

Radiofrequency ablation for Dysplasia

Before Ablation



After Ablation



24

Treatment Plan

Superficial (T1)
- Endoscopic Therapy

Localized (T1b/T2)
- Surgery (esophagectomy)

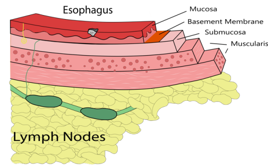
Locally-advanced (T3M0)
- Chemo±Radiation →Surgery

Metastatic (M1)
- Chemotherapy

25

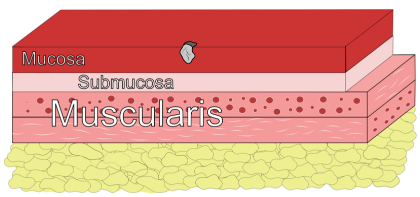
Superficial Cancers = T1a

Treatment is often with by endoscopic resection without the need for surgery.



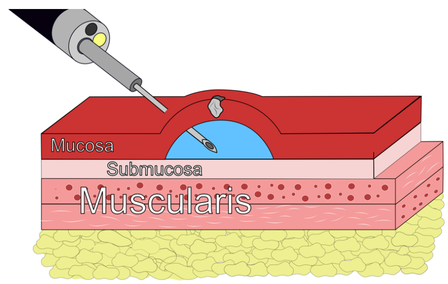
26

Endoscopic Resection



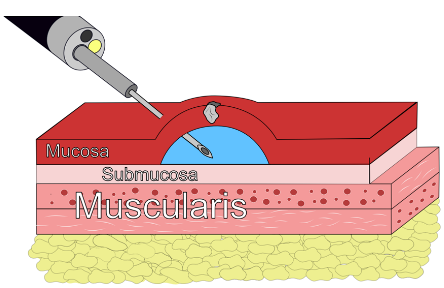
27

Tumor Lift



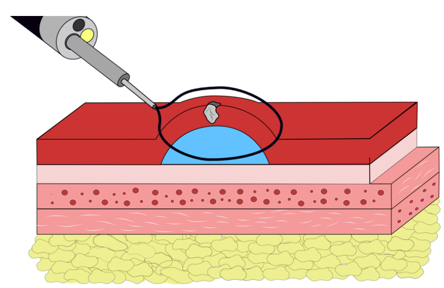
28

Tumor Lift



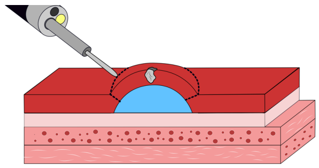
29

Tumor Removal - Resection



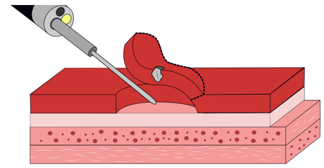
30

Tumor Removal - Dissection



31

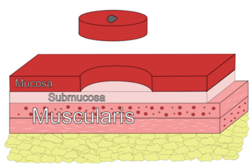
Tumor Removal - Dissection



32

Endoscopic Resection - Favorable

- Clear edge margin *AND*
- Clear deep margin *AND*
- Tumor slow-growing under microscope



33

Endoscopic Resection - Favorable

- Clear edge margin *AND*
- Clear deep margin *AND*
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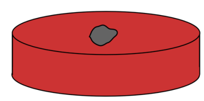
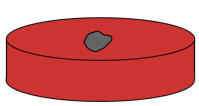


34

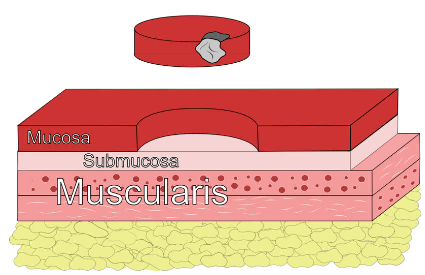
Endoscopic Resection - Favorable

- May be the only treatment required
- Requires endoscopic surveillance



35

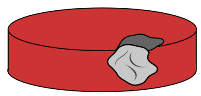
Endoscopic Resection - Unfavorable



36

Endoscopic Resection - Unfavorable

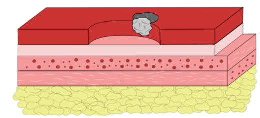
- Tumor at edge margin
  OR
- Tumor at deep margin
  OR
- Tumor rapidly-growing under microscope



37

Endoscopic Resection - Unfavorable

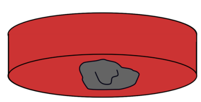
Esophagectomy (surgery) is standard recommendation



38

Endoscopic Resection - Positive Deep Margin

- Tumor at edge margin
  OR
- **Tumor at deep margin**
  OR
- Tumor rapidly-growing under microscope



39

Endoscopic Resection - Positive Deep Margin

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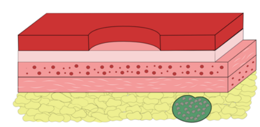

40

Endoscopic Resection - Positive Deep Margin

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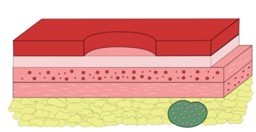
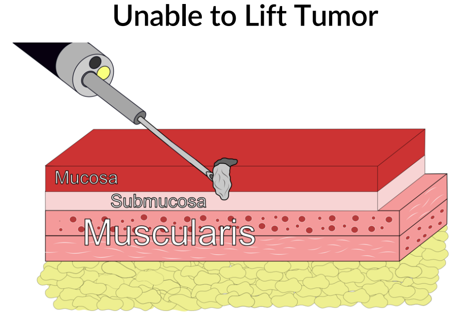


41

Unable to Lift Tumor



42